Review #4

1) a) $\int \tan^4(2x)\,dx$

$= \int \tan^2(2x)\tan^2(2x)\,dx$

$= \int \tan^2(2x)\left(\sec^2(2x)-1\right)dx$

$= \int \tan^2(2x)\sec^2(2x)\,dx - \int \tan^2(2x)$

$= \int u^2 \cancel{\sec^2(2x)} \frac{du}{2\cancel{\sec^2(2x)}} - \int \sec^2(2x)-1\,dx$

$= \boxed{\frac{u^3}{6} - \frac{\tan(2x)}{2} + x + C}$

$u = \tan(2x)$

$du = 2\sec^2(2x)\,dx$

$dx = \frac{du}{2\sec^2 2x}$

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b) $\int \frac{4\sin x}{\cos^3 x}\,dx$

$= 4\int \frac{\sin}{\cos^2 x \cos x}\,dx = \int \frac{\tan}{\cos^4}\,dx = \int \tan x \sec^2 x$

$u = \tan$

$du = \sec^2 x\,dx$

$\int \tan x \sec^2 x\,dx = \int u \frac{\sec^2 du}{\sec^2 x} = \int u\,du = \frac{u^2}{2}$

$= \boxed{\frac{\tan^2 x}{2} + C}$

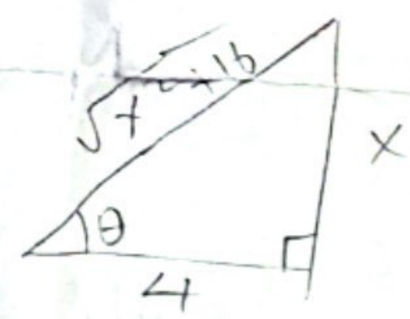
Evaluate the integral using trigonometric substitution

a) $\int \frac{x^3}{\sqrt{x^2+16}}\,dx$ $\quad x = 4\tan(\theta)$, sketch a label the right triangle

$\sqrt{x^2+16} = \sqrt{(4\tan\theta)^2+16} = \sqrt{16\tan^2\theta+16}$

$= \sqrt{16(\tan^2+1)}$

$= 4\sec\theta$



$\int \frac{x^3}{\sqrt{x^2+16}} = \frac{64\tan^3\theta}{4\sec\theta}\,dx$

$dx = 4\sec^2\theta\,d\theta \Rightarrow \int \frac{\tan^3\theta}{\sec\theta} 4\sec^2\theta\,d\theta$

$= 64\int \tan^2\theta \tan\theta\sec\theta\,d\theta = 64\int (\sec^2\theta - 1)\tan\theta\sec\theta\,d\theta$

$u = \sec\theta$

$du = \sec(\theta)\tan\theta\,d\theta \Rightarrow 64\int (u^2-1)\frac{\tan\theta\sec\theta\,du}{\tan\theta\sec\theta} = 64\int u^2 - 1\,du$

$= 64\left(\frac{u^3}{3} - u\right) = 64\left(\frac{\sec^3\theta}{3} - \sec\theta\right) + C$

$\theta = \tan^{-1}(x/4)$

$= \boxed{64\left(\frac{\sec^3(\tan^{-1}(x/4))}{3} - \sec(\tan^{-1}(\frac{x}{4}))\right) + C}$

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$\sqrt{1-81x^2}\,dx$

use $9x = \sin\theta$ $\qquad dx = \frac{\cos\theta}{9}\,d\theta$

$$\sqrt{1-\sin^2\theta}\,dx = \int \sqrt{1-\sin^2\theta}\,\frac{\cos\theta}{9}\,d\theta$$

$$= \frac{1}{9}\int \sqrt{\cos^2\theta}\,\cos\theta\,d\theta = \frac{1}{9}\int \cos^2\theta\,d\theta$$

$$= \frac{1}{9}\int \frac{(1+\cos 2\theta)}{2}\,d\theta = \frac{1}{18}\left(\int 1\,d\theta + \int \cos 2\theta\,d\theta\right)$$

$$= \frac{\theta}{18} + \frac{\sin 2\theta}{36} + C = \frac{1}{18}\left[\theta + \sin\theta\cos\theta\right] + C$$

$\theta = \sin^{-1}(9x)$, $\int \sqrt{1-81x^2}\,dx = \boxed{\frac{1}{18}\left(\sin^{-1}(9x) + 9x\cos(\sin^{-1}(9x))\right) + C}$

c) $\int \frac{\sqrt{x^2-1}}{x}$ $\qquad x = \sec\theta$ $\qquad dx = \tan\theta\sec\theta\,d\theta$

$\sqrt{x^2-1} = \sqrt{(\sec^2 - 1)} = \tan\theta$ $\qquad \int \frac{\tan\theta}{\sec\theta}\tan\theta\sec\theta\,d\theta$

$$= \int \tan^2\theta\,d\theta = \int (\sec^2 - 1)\,d\theta$$

$$= \int \sec^2\theta\,d\theta - \int 1\,d\theta$$

$$= \tan\theta - \theta + C$$

$\theta = \sec^{-1}(x)$ $\qquad = \boxed{\tan(\sec^{-1}(x)) - \sec^{-1}(x) + C}$

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